Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
3	(a)			**Indicative content:** **<u>Experimental set-up</u>** Source clear in diagram or description Detector clear in diagram or description Distance clear in diagram or description. Blocking alpha and beta [with aluminium]. **<u>Data collection</u>** Vary distance. Measure counts for set time / long time / repeats Check for background With no source present **<u>Analysis</u>** Adjust for the background intensity (or nothing if they have said that the source is a strong source / lead container). Correct plot chosen e.g. intensity vs r^{-2}, or $(\text{count})^{-1/2}$ vs r $\ln(\text{intensity or count})$ vs $\ln r$ Correct expected shape i.e. straight line Added detail e.g. error bars, intercept etc	6			6		6

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
3				**5-6 marks** A comprehensive account of all 3 areas is provided. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.* **3-4 marks** A comprehensive account is given for 2 out of the 3 areas i.e. experimental set up, data collection and analysis or a limited account of all 3 areas is provided. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.* **1-2 marks** A comprehensive account is given for 1 out of the 3 areas i.e. experimental set up, data collection and analysis or a limited account of 1 or 2 areas is provided. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.* **0 marks** *No attempt made or no response worthy of credit.*						
	(b)	(i)	I	Substitution: $E = hf = h\frac{c}{\lambda}$ $E = (6.63 \times 10^{-34})\left(\frac{3\times10^{8}}{1\times10^{-12}}\right)(1)$ $= 2.0 \times 10^{-13}[\mathrm{J}](1)(\text{or } 1.24\ \mathrm{MeV})$	1	1		2	2	
			II	$E = (6.63 \times 10^{-34})(1.8 \times 10^{9}) = 1.2 \times 10^{-24}[\mathrm{J}]$		1		1	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		**For the 1st mark:** Lots of further research / data collected **For the 2nd mark either of:** – Brain cancer vs mobile phone usage i.e. targeting brain cancer victims or high mobile phone usage – Carry out statistical analysis e.g. look for correlations between usage (in phone bills) and cancer occurrence. Don't accept carry out experiments on humans or any reference to ionisation or testing what comes out of mobile phones			2	2		
				Question 3 total	**7**	**2**	**2**	**11**	**3**	**6**